12 - Slotted ALOHA Performance

Let:

- p = probability that one node transmits in a slot
- n = number of nodes
- $1-p$ = probability that one node does not transmit in a slot

For one specific user to transmit successfully:

- that user must transmit, which happens with probability p
- the remaining $n-1$ users must stay silent, each with probability $(1-p)$

So,

$$\begin{aligned} P(\text{one specific user succeeds}) &= p(1-p)(1-p)\cdots(1-p) \\ &= p(1-p)^{n-1} \end{aligned}$$

Probability That Any User Succeeds

Any one of the n users could be the successful user, so

$$\begin{aligned} P_s &= p(1-p)^{n-1} + p(1-p)^{n-1} + \cdots + p(1-p)^{n-1} \\ &= n\left[p(1-p)^{n-1}\right] \end{aligned}$$

Hence, the success probability of slotted ALOHA is

$$P_s = np(1-p)^{n-1}$$

Maximizing the Success Probability

We want to choose p to maximize P_s.

$$P_s = np(1-p)^{n-1}$$

Differentiate with respect to p:

$$\begin{aligned}\frac{dP_s}{dp} &= n\left[(1-p)^{n-1} + p(n-1)(1-p)^{n-2}(-1)\right] \\ &= n(1-p)^{n-1} - np(n-1)(1-p)^{n-2}\end{aligned}$$

Set the derivative equal to zero:

$$\begin{aligned}0 &= n(1-p)^{n-1} - np(n-1)(1-p)^{n-2} \\ &= n(1-p)^{n-2}\left[(1-p) - p(n-1)\right]\end{aligned}$$

So,

$$\begin{aligned}(1-p) - p(n-1) &= 0 \\ 1 - p - pn + p &= 0 \\ 1 - pn &= 0 \\ pn &= 1 \\ p^* &= \frac{1}{n}\end{aligned}$$

Thus, the value of p that maximizes the success probability is

$$p^* = \frac{1}{n}$$

Substitute $p^* = \frac{1}{n}$ into P_s:

$$\begin{aligned}P_s &= np^*(1-p^*)^{n-1} \\ &= n\left(\frac{1}{n}\right)\left(1 - \frac{1}{n}\right)^{n-1} \\ &= \left(1 - \frac{1}{n}\right)^{n-1}\end{aligned}$$

This is the probability of success for the whole system when each user transmits with the optimal probability p^*.

Capacity

Given:

$$\text{Capacity of the network} = \frac{P_s \cdot L}{T_{\text{slot}}} \text{ bits/sec}$$

where:

- L = number of bits transmitted in one successful slot
- T_{slot} = time duration of one slot

if

$$R = \frac{L}{T_{\text{slot}}}$$

is the transmission rate in bits/sec, then

$$\text{Capacity of the network} = P_s \cdot R$$

Capacity per user is

$$\text{Capacity per user} = \frac{P_s \cdot R}{n}$$

Here, P_s is a success probability per slot, so to convert it to bits/sec we multiply by the number of bits in a successful slot and divide by the slot duration.

Behavior as the Number of Users Grows

Now let $n \to \infty$:

$$\begin{aligned} P_s &= \left(1 - \frac{1}{n}\right)^{n-1} \\ &= \left(\frac{n-1}{n}\right)^{n-1} \end{aligned}$$

Recall:

Using the binomial expansion,

$$(a+x)^t = \sum_{k=0}^{t} \binom{t}{k} a^{t-k} x^k$$

Apply this to $\left(1 - \frac{1}{n}\right)^{n-1}$:

$$\begin{aligned} \left(1 - \frac{1}{n}\right)^{n-1} &= \sum_{k=0}^{n-1} \binom{n-1}{k} (1)^{n-1-k} \left(-\frac{1}{n}\right)^k \\ &= \sum_{k=0}^{n-1} \binom{n-1}{k} \frac{(-1)^k}{n^k} \\ &= \sum_{k=0}^{n-1} \frac{(n-1)!}{k!(n-1-k)!} \frac{(-1)^k}{n^k} \end{aligned}$$

Now take the limit:

$$\begin{aligned} \lim_{n\to\infty} P_s &= \lim_{n\to\infty} \sum_{k=0}^{n-1} \frac{(n-1)!}{k!(n-1-k)!} \frac{(-1)^k}{n^k} \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \end{aligned}$$

because for fixed k,

$$\begin{aligned} \lim_{n\to\infty} \frac{(n-1)!}{(n-1-k)!\, n^k} &= \lim_{n\to\infty} \frac{(n-1)(n-2)\cdots(n-k)}{n^k} \\ &= 1 \end{aligned}$$

Therefore,

$$\begin{aligned} \lim_{n\to\infty} P_s &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \\ &= e^{-1} \\ &\approx 0.37 \end{aligned}$$

Result

The maximum throughput of slotted ALOHA, as the number of users becomes very large, is

$$e^{-1} \approx 0.37$$

So the efficiency of slotted ALOHA is about 37%.

Pure ALOHA Performance

In Pure ALOHA, a node transmits immediately when it has a frame. There are no synchronized slots.

The key difference is the vulnerable period:

- in Slotted ALOHA, a frame only collides if another node transmits in the same slot
- in Pure ALOHA, a frame can collide with another frame that starts within one packet time before or after it

So the vulnerable period is 2 packet times instead of `1`.

Success Probability for One Specific User

For one specific user to succeed in Pure ALOHA:

- that user must transmit, which happens with probability p
- no other node may transmit in the packet-time interval before it
- no other node may transmit in the packet-time interval after it

So,

$$\begin{aligned} P(\text{one specific user succeeds}) &= p(1-p)^{n-1}(1-p)^{n-1} \\ &= p(1-p)^{2(n-1)} \end{aligned}$$

Success Probability for the Whole System

Any one of the n users could be the successful user, so

$$P_s = np(1-p)^{2(n-1)}$$

Maximizing the Pure ALOHA Success Probability

We want to maximize

$$P_s = np(1-p)^{2(n-1)}$$

Differentiate with respect to p:

$$\frac{dP_s}{dp} = n\left[(1-p)^{2(n-1)} - 2p(n-1)(1-p)^{2(n-1)-1}\right]$$

Set the derivative equal to zero and factor:

$$0 = n(1-p)^{2n-3}\left[(1-p) - 2p(n-1)\right]$$

So,

$$\begin{aligned} (1-p) - 2p(n-1) &= 0 \\ 1 - p - 2pn + 2p &= 0 \\ 1 + p - 2pn &= 0 \end{aligned}$$

For large n, this gives approximately

$$p^* \approx \frac{1}{2n}$$

Behavior as the Number of Users Grows

Substitute the approximate optimum $p^* = \frac{1}{2n}$:

$$\begin{aligned} P_s &= n\left(\frac{1}{2n}\right)\left(1-\frac{1}{2n}\right)^{2(n-1)} \\ &= \frac{1}{2}\left(1-\frac{1}{2n}\right)^{2(n-1)} \end{aligned}$$

Now let $n \to \infty$:

$$\begin{aligned} \lim_{n\to\infty} P_s &= \frac{1}{2}e^{-1} \\ &= \frac{1}{2e} \\ &\approx 0.18 \end{aligned}$$

So the efficiency of Pure ALOHA is about 18%, which is worse than Slotted ALOHA because the vulnerable period is twice as large.